

Project: Small Network at Tech Institute- Critical Issues and Planning a Migration

CNET – 301-001

Network Design and Test

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ABSTRACT

In this project the migration is to be planned of the current network which has some vulnerabilities in it. Basically, there are 4 campuses – A, B, C, D and the network, which is now out of date, grew unintentionally and without any technological thinking over the course of the year. All Campus User Groups that have been frequently polled have consistently voiced their frustration with the frustratingly slow response from network services including database searches, printing, and internet browsing as well as with the slow logins and downloads. For internet connectivity, each campus has a unique deal with an ISP. In accordance with this issue, we now plan a migration of the old network to a new advanced one which eliminates these problems. Therefore, overall, in this project we plan to redesign the whole network in such a way that it can later accommodate future growth of users by 20% per year at all the Campuses. Furthermore, we will be upgrading the hardware which will have high fault tolerance rate, implement firewall, remote login etc.

INTRODUCTION

In this project, we plan a migration over a period of two years of the current network which is outdated to an advanced network which is scalable, secure, flexible, and reliable. Thus, the four-campus Tech Institute intends to replace its current network infrastructure with a much better one that has high speed and appropriate placements of modern gear. These improvements will offer reliable hardware and software with lower failure rates. Campuses of the Tech Institute have total 4 campuses. These four campuses overall have 28 classrooms with total of 7,450 students including 50 support staff, 20 Managers and 50 instructors. Each classroom has 32 P4 1 GHz PCs having 1 GB RAM and legacy NIC Cards. Therefore, in general, we intend to restructure the entire network as part of this project so that it can subsequently handle potential user growth of 20% per year at all Campuses. Additionally, we will upgrade the hardware to include features like a firewall, remote login, and high failure tolerance with also making the campus printing friendly with the budget of \$30,0000 per academic year.

OBJECTIVES

1. To redesign the entire network using enterprise network ideas such as the Core layer, Distribution layer, and Access layer. Create a system bandwidth plan for the users to account for expected 20 percent annual growth across all campuses.
2. To provide performance and fault tolerance as deemed necessary at all Campuses, upgrade a number of the server hardware platforms.
3. To install an Internet firewall to offer protection.
4. To Install a more powerful uninterruptible power supply, or UPS, system in order to protect servers and other critical infrastructure equipment from power outages.
5. To provide 400 employees with safe remote login connections so they can operate remotely from locations other than their offices. In this situation, choose the technologies that are appropriate.
6. To estimate the number of Wi-Fi access points that will be required at each campus.

PROJECT SUMMARY

Tech Institute, a four-campus university, seeks to improve every aspect of its network architecture by switching to a faster network with properly arranged modern hardware. With these updates, hardware and software will be more reliable and efficient. It is also possible to enhance qualities like scalability, flexibility, dependability, performance, and security. The following characteristics apply to Tech Institute campuses:

- 7450 pupils in all, 50 members of support staff, 50 teachers, and 20 managers.
- **Campus-A** exclusively houses the School of Media and includes 4 classes (500 students)
- **Campus-B** is home to three schools: Business (2000 students), Engineering (3000 students), and Health Sciences (150 students), and it includes 12 classrooms.
- **Campus-C** has 8 classrooms, 350 students enrolled in the School of General Arts, and IT services (support staff of 20).
- **Campus-D** houses the School of Aviation and has four classes (100 students).

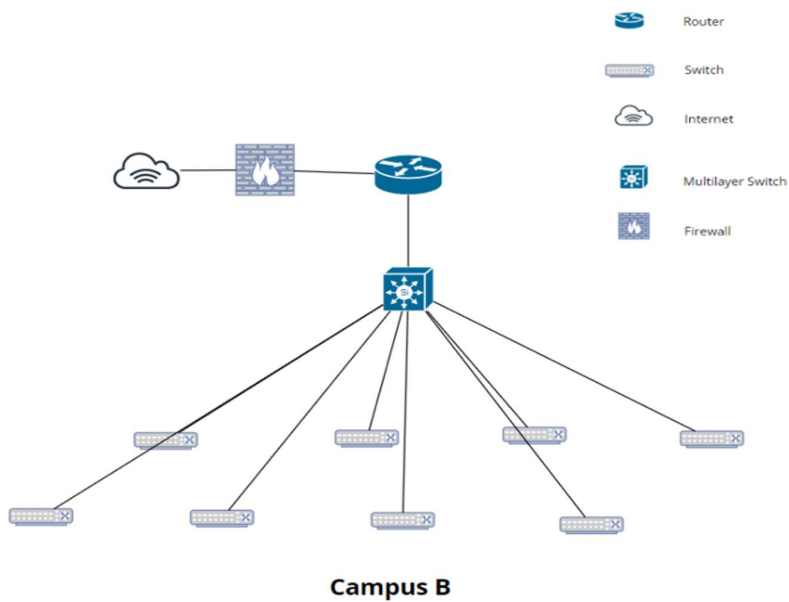
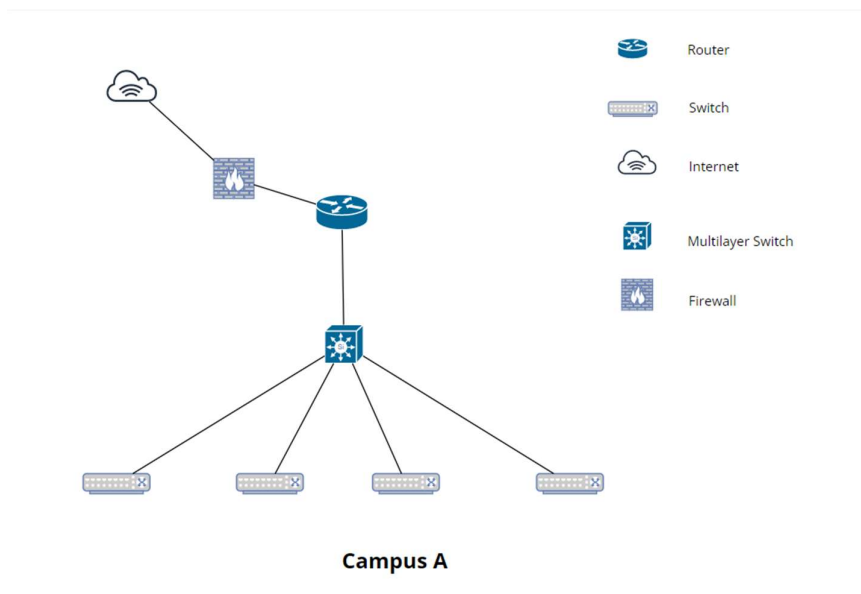
All Campus User Groups that have been frequently polled have consistently voiced their frustration with the extremely slow response from network services including database searches, printing, and internet browsing as well as with the slow logins and downloads. Therefore, working for improved printer accessibility across all campuses is also one of our goals.

Furthermore, considering its technical aspects, 32 P4 1GHz PCs with 1 GB of RAM and legacy NIC cards are available in each classroom. The RAM on each PC was increased to 2 GB. Specifically, Campus A has 128 computers, while Campus B has 384 computers, Campus C has 256 computers, whereas Campus D has 128. Moreover, six servers make up the Server Farm, which is situated on Campus-C. These servers each have a 1 GHz Intel XEON processor, 1 GB of RAM, and a variety of older NICs.

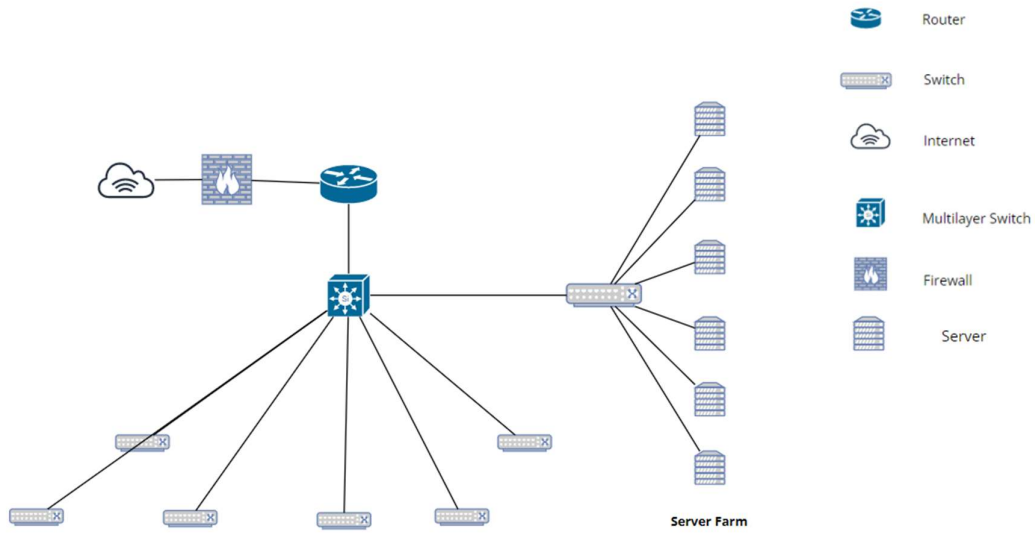
We must implement this overall in a budget of **\$30,000** per year. So, the upgradation cost for 2 years equals to **\$60,000**.

NETWORK DESIGNS

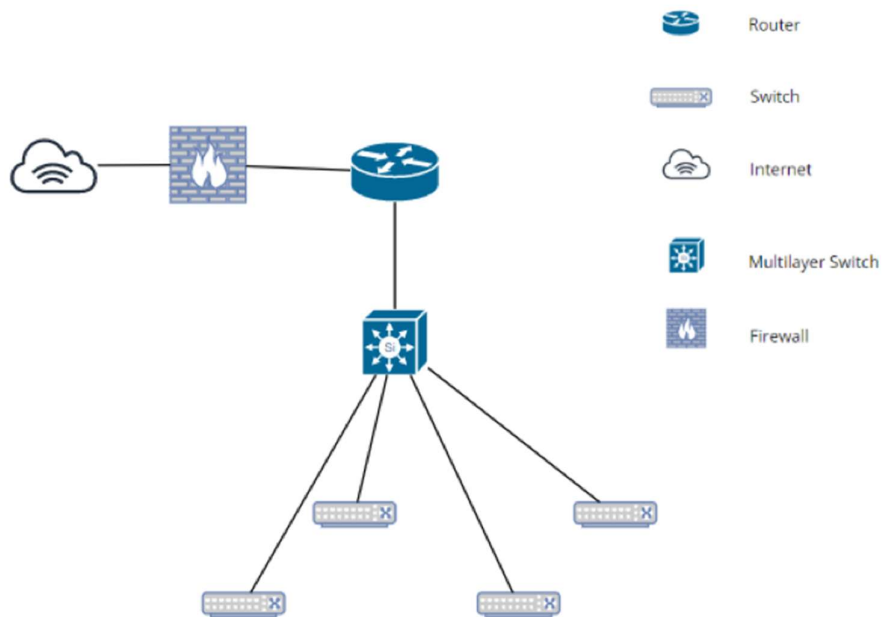
Stage I:



STAGE II:



Campus C



Campus D

IP ADDRESS DISTRIBUTION

Campus	IP range	Subnet Mask
Campus A	172.16.2.1 – 172.16.2.254	/23
Campus B	172.16.3.1 – 172.16.5.254	/23
Campus C	172.16.6.1 – 172.16.7.254	/24
Campus D	172.16.8.1 – 172.16.8.254	/24

Switched VLAN	VLAN Names	Network Address	Subnet Mask	Default gateway
Campus A	VLAN 1	172.16.2.0	255.255.255.192	172.16.2.1
Campus A	VLAN 2	172.16.2.64	255.255.255.192	172.16.2.65
Campus A	VLAN 3	172.16.2.128	255.255.255.192	172.16.2.129
Campus A	VLAN 4	172.16.2.192	255.255.255.192	172.16.2.193
Campus A - D	A - D	172.16.11.0	255.255.255.240	172.16.11.1
Campus B	VLAN 1	172.16.3.0	255.255.255.192	172.16.3.1
Campus B	VLAN 2	172.16.3.64	255.255.255.192	172.16.3.65
Campus B	VLAN 3	172.16.3.128	255.255.255.192	172.16.3.129
Campus B	VLAN 4	172.16.3.192	255.255.255.192	172.16.3.193
Campus B	VLAN 5	172.16.4.0	255.255.255.192	172.16.4.1
Campus B	VLAN 6	172.16.4.64	255.255.255.192	172.16.4.65
Campus B	VLAN 7	172.16.4.128	255.255.255.192	172.16.4.129

Campus B	VLAN 8	172.16.4.192	255.255.255.192	172.16.4.193
Campus B	VLAN 9	172.16.5.0	255.255.255.192	172.16.5.1
Campus B	VLAN 10	172.16.5.64	255.255.255.192	172.16.5.65
Campus B	VLAN 11	172.16.5.128	255.255.255.192	172.16.5.129
Campus B	VLAN 12	172.16.5.192	255.255.255.192	172.16.5.193
Campus B - D	B - D	172.16.11.16	255.255.255.240	172.16.11.17
Campus C	VLAN 1	172.16.6.0	255.255.255.192	172.16.6.1
Campus C	VLAN 2	172.16.6.64	255.255.255.192	172.16.6.65
Campus C	VLAN 3	172.16.6.128	255.255.255.192	172.16.6.129
Campus C	VLAN 4	172.16.6.192	255.255.255.192	172.16.6.193
Campus C	VLAN 5	172.16.7.0	255.255.255.192	172.16.7.1
Campus C	VLAN 6	172.16.7.64	255.255.255.192	172.16.7.65
Campus C	VLAN 7	172.16.7.128	255.255.255.192	172.16.7.129
Campus C	VLAN 8	172.16.7.192	255.255.255.192	172.16.7.193
Campus C- D	C - D	172.16.11.32	255.255.255.240	172.16.11.33
Campus D	VLAN 1	172.16.8.0	255.255.255.192	172.16.8.1
Campus D	VLAN 2	172.16.8.64	255.255.255.192	172.16.8.65
Campus D	VLAN 3	172.16.8.128	255.255.255.192	172.16.8.129
Campus D	VLAN 4	172.16.8.192	255.255.255.192	172.16.8.193
Campus D - D	D -D	172.16.11.48	255.255.255.240	172.16.11.49
Server Farm	S F	172.16.22.0	255.255.255.224	172.16.22.1

ACCESS POINT DISTRIBUTION TABLE

VLAN NAMES	NETWORK ADDRESS	SUBNET MASK	Default Gateway
WAP - A	192.168.7.0	255.255.255.192	192.168.7.1
WAP - B	192.168.7.64	255.255.255.192	192.168.7.65
WAP - C	192.168.7.128	255.255.255.192	192.168.7.129
WAP - D	192.168.7.192	255.255.255.192	192.168.7.193

Campus A Configuration:

Interface	IP Address	Subnet Mask
Firewall		
G0/0 connected to the router	10.10.10.10	255.255.255.224
G0/1 connected to cloud	10.10.10.11	255.255.255.224
Router		
G0/0	10.10.10.9	255.255.255.224
G0/1 connected to switch	10.10.10.8	255.255.255.224
Switch		
G0/1 connected to router	10.10.10.7	255.255.255.224

Campus B Configuration:

Interface	IP Address	Subnet Mask
Firewall		
G0/0 connected to the router	10.10.10.15	255.255.255.224
G0/1 connected to cloud	10.10.10.14	255.255.255.224
Router		
G0/0	10.10.10.13	255.255.255.224
G0/1 connected to switch	10.10.10.12	255.255.255.224
Switch		
G0/1 connected to router	10.10.10.16	255.255.255.224

Campus C Configuration:

Interface	IP Address	Subnet Mask
Firewall		
G0/0 connected to the router	10.10.10.21	255.255.255.224
G0/1 connected to cloud	10.10.10.20	255.255.255.224
Router		
G0/0	10.10.10.19	255.255.255.224
G0/1 connected to switch	10.10.10.18	255.255.255.224
Switch		
G0/1 connected to router	10.10.10.17	255.255.255.224

Campus D Configuration:

Interface	IP Address	Subnet Mask
Firewall		
G0/0 connected to the router	10.10.10.26	255.255.255.224
G0/1 connected to cloud	10.10.10.25	255.255.255.224
Router		
G0/0	10.10.10.24	255.255.255.224
G0/1 connected to switch	10.10.10.23	255.255.255.224
Switch		
G0/1 connected to router	10.10.10.22	255.255.255.224

Hardware Details:

UPS: APC- SYH4K6RMT

89% Full load efficiency

208V Nominal I/O Voltage

Output Power Capacity 4.2K Watts/6.0 KVA

3 Battery modules

Layer 3 Switch: Cisco WS-c2960+48PST-S

- ❑ 48 Fast Ethernet Ports
- ❑ Flash memory 64MB
- ❑ 64 MB RAM
- ❑ 6.5 Mpps throughput
- ❑ 2*10/100/1000 uplinks and 2 SFP uplinks

Wireless Access Point: Cisco Catalyst C9200L-24P-4G-E

- ❑ 24 POE+ Ethernet ports
- ❑ 4 Gigabit SFP uplink ports
- ❑ 4GB flash memory
- ❑ 2 GB DRAM
- ❑ 80 Gbps stacking bandwidth

Firewall: ASA5516-FPWR-K9

- ❑ 850 Mbps maximum throughput

- ☐ Unlimited nodes and users
- ☐ 8 GB Flash memory
- ☐ 8 GB memory

Additional details:

Router: Cisco 4000 Router ISR4331(3GE,2NIM,1SM,4G FLASH,4G DRAM, IP Base)

- 2 RJ45 ports
- 2 SFP ports
- 2 NIM slots
- Included OIR (all I/O modules)
- Total 3 units of WAN or LAN 10/100/1000 ports
- 2Gb/s + Performance
- 100-300 Mb/s throughput
- 1 external USB 2.0 slot
- 1 USB console port and 1 serial console port
- Default 4Gb Flash memory / 16Gb Maximum
- 100 to 240 VAC auto ranging AC input voltage
- 47 to 63 Hz AC input frequency
- Maximum power – 250 AC supply

<https://www.router-switch.com/cisco-isr4331-k9-p-16544.html>



Layer 3 Switch: WS-C2960+48TC-S Catalyst 2960 Plus Switch

- 48 Fast Ethernet Ports
- 2 SFP uplink interfaces
- LAN Lite Cisco IOS Software
- Flash memory 64MB
- 128 MB DRAM
- 16 Gbps Forwarding Bandwidth
- Maximum 64 VLANs
- 9000 bytes Maximum transmission unit (MTU)
- 623 thousand of hours MTBF
- 100 to 240 VAC voltage
- 0.6-0.3 Amperes Current
- 50-60 Hz Frequency
- 0.04 Power Rating kVA
- 3A at 12V DC input voltages (RPS input)
- Environmentally friendly with RoHS

<https://www.router-switch.com/ws-c2960-48tc-s-p-5262.html>



Switch: SG220-26P - Cisco 220 Series Smart Switches

- 128 MB RAM
- 26 switch ports
- 24 * 10/100/1000 + 2 combo Gigabit SFP
- PoE and PoE+
- 38.69 Mbps switch capacity
- 8192 entries of MAC address table
- Secure Shell (SSH), Radius and Tacacs+ authentication
- 128 Mb RAM
- 32 MB Flash memory
- 50 RJ45 Gigabit Ethernet ports
- Internal power supply
- Included rack mounting kit
- 100-240 V, 50-60 Hz voltage required
- 0-50 Degree Celsius operating temperature

<https://www.router-switch.com/sg220-26p.html>



Wireless LAN Controller: C1-AIR-CT3504-K9 - Cisco WLAN Controller

- 4 Gbps throughput
- 4*10/100/1000 ports
- 10/100/1000 RJ-45 console ports
- CIDS Routing Protocol
- PoE supported
- Maximum access points are 150
- Maximum number of clients are 3000
- VLANs supported are 4096
- 120/230 V AC required voltage
- 32 F to 104 F operating Temperature

<https://www.router-switch.com/c1-air-ct3504-k9.html>



Firewall: ASA5545-K9 Cisco ASA 5500 Series Firewall

- 2Gbps maximum throughput
- 900 Mbps IPS throughput
- 2500 IPsec VPN peers
- 5000 Cloud Web Security users

- 1 Gbps Next Generation Multiprotocol Throughput (Multiprotocol)
- Unlimited nodes and users
- 30000 New connections per second
- 2 USB 2.0 ports
- 8 GB Flash memory
- 12 GB memory
- Multibus architecture
- 100 to 240 VAC voltage Range

<https://www.router-switch.com/asa5545-k9-p-4623.html>



Wireless Access Point: C9300L-24T-4G-A - Cisco Catalyst 9300L

- 24 POE+ Ethernet ports
- 4 Gigabit SFP uplink ports
- Ipv4 routing entries 32,000 (24,000 direct routes and 8000 indirect routes)
- 16GB flash memory
- 8 GB DRAM
- 320 Gbps stacking bandwidth

<https://www.router-switch.com/c9300l-24t-4g-a.html>



UPS: APC Symmetra RM 2kVA Scalable to 6kVA N+1 208/240V w/ 208 to 120V Step-Down Transformer (4) L5-20R

- 88% full load efficiency
- 208 V nominal I/O voltage
- 1400 W rated power
- Maximum 4200 W configurable power
- NEMA L6-30P 30 A Standard Plug
- 10 Unit racks

<https://www.apc.com/ca/en/product/SYH2K6RMT-TF3/apc-symmetra-rm-2kva-scalable-to-6kva-n+1-208-240v-w-208-to-120v-stepdown-transformer-4-l520r/>



Wireless LAN Controller: Cisco AIR-CT2504-5-K9

- ☒ 1 Gbps throughput
- ☐ 4*10/100/1000 ports
- ☒ 10/100/1000 RJ-45 console ports
- ☒ Maximum access points are 75

Switch: Cisco SG220-50-K9-NA

- ☒ 128 MB memory
- ☒ 32 MB Flash memory
- ☒ 50 RJ45 Gigabit Ethernet ports

Router: Cisco ISR4331/K9

- ☒ RJ45/SFP ports
- ☒ 2 NIM slots
- ☒ 2Gb/s + Performance
- ☒ 100 Mb/s throughput
- ☒ 4GB Flash memory

COST ESTIMATION**Campus A**

Devices	Type	Quantity	Cost	Total
ASA5516-FPWR- K9	Firewall	1	\$1300	\$1300
Cisco WS-c2960+48PST-S	Switch	1	\$500	\$500
HP DeskJet Plus 4155	Printer	6	\$140	\$840
Cisco SG220-50- K9-NA	Switch	4	\$400	\$1600
APC SYH4K6RMT -	UPS	1	\$4100	\$4100
Cisco AIR-CT2504-5-K9	WLAN Controller	1	\$700	\$700
Cisco ISR4331/K9	Router	1	\$800	\$800
Cisco Catalyst C9200L-24P-4G-E	Wireless AP	4	\$700	\$2800
			TOTAL	CDN \$12640

Campus -B

Devices	Type	Quantity	Cost	Total
CiscoCatalyst C9200L-24P-4G-E	Wireless AP	4	\$700	\$2800
APC SYH4K6RMT	UPS	1	\$4100	\$4100
HP DeskJet Plus 4155	Printer	6	\$140	\$840
ASA5516-FPWR-K9	Firewall	1	\$1300	\$1300
Cisco SG220-50-K9-NA	Switch	9	\$400	\$3600
Cisco ISR4331/K9	Router	1	\$800	\$800
Cisco WS-c2960+48PST-S	Switch	1	\$500	\$500
Cisco AIR-CT2504- 5-K9	WLAN Controller	1	\$700	\$700
Total				CDN \$14640

Total cost for Campus A=CDN \$12640

Total cost for Campus B=CDN \$14640

The total cost of Campus A and B up-gradation for the first year=CDN \$27280

CAMPUS C

Devices	Type	Quantity	Cost	Total
ASA5516-FPWR- K9	Firewall	1	\$1300	\$1300
Cisco Catalyst C9200L-24P-4G-E	Wireless AP	3	\$700	\$2100
Cisco SG220-50- K9-NA	Switch	6	\$400	\$2400
APC - SYH4K6RMT	UPS	1	\$4100	\$4100
Cisco WS-c2960+48PST-S	Switch	1	\$500	\$500
HP DeskJet Plus 4155	Printer	4	\$140	\$560
Cisco ISR4331/K9	Router	1	\$800	\$800
Cisco AIR-CT2504-5-K9	WLAN Controller	1	\$700	\$700
Total				CDN \$12460

CAMPUS D

Devices	Type	Quantity	Cost	Total
ASA5516-FPWR- K9	Firewall	1	\$1300	\$1300
Cisco SG220-50- K9-NA	Switch	4	\$400	\$1600
APC SYH4K6RMT	UPS	1	\$4100	\$4100
Cisco AIR-CT2504-5-K9	WLAN Controller	1	\$700	\$700
HP DeskJet Plus 4155	Printer	1	\$140	\$140
Cisco WS-c2960+48PST-S	Switch	1	\$500	\$500
Cisco Catalyst C9200L-24P-4G-E	Wireless AP	2	\$700	\$1400
Cisco ISR4331/K9	Router	1	\$800	\$800
			TOTAL	\$10540

IMPLEMENTATION PLAN

For more flexibility and performance, networks must be upgraded. For Campus A and B, we intend to begin the first stage of network upgrading on September 10, 2022. For Campus C and D, we intend to begin the second stage of network upgrading on September 10, 2023. The main advantage of choosing this day is the fact that less college resources are utilized. Therefore, now is the ideal time to upgrade without too much hassle. Through the college mail system, all employees, including students and other staff, are informed of the upgrades or maintenance.

In the evenings and on weekends, we test all the new sets and apparatus. All new equipment is correctly connected before the implementation date. Additionally, campus off days like vacations and semester breaks are the perfect times for these implementations.

CONCLUSION

We can improve the network in two stages while staying within the budget. For two years, the expected total cost is CDN \$50280. The redesign and application of the suggested network topology will successfully address all network-related problems.